Cantor's theorem: A is strictly smaller than its powerset

The Open Logic Project contributors

openlogic:sets-functions-relations/size-of-sets:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-sets-functions-relations-size-of-sets/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: \oliflabeldef{sfr:siz:nen:sec}{slow branch}{fast branch} renders the fast reductio under the current preamble semantics (else group); both branches are complete author arguments. The editorial note inside the proof about which branch to present is the author's own and rendered as such.

Editorial note. The label thm:cantor names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own thm environment in the unit source.

Allowed standard prerequisites (editorial).

- Extensionality. Same unit, set extensionality (standing axiom/definition of the chapter). Injectivity of x -> {x}: singletons of distinct sets are distinct.

Source reference. Original reference sec: sources/olp-sets-functions-relations-size-of-sets/pinned-original-source.tex, source lines 469–469. Named citation invoked at source line(s) 469: the author's marker \olref[nen-alt]{sec} resolves inside the full Open Logic Project to another unit (not staged in this corpus); the cited fact is used as a standard result and every step of the counted proof built on it is retained verbatim in the delivered body.

Theorem 0.1 (Cantor)**.** *$A \prec \wp(A)$, for any set A.*

Proof. The map $f(x) = \{x\}$ is injection $f\colon A \to \wp(A)$, since if $x \neq y$, then also $\{x\} \neq \{y\}$ by extensionality, and so $f(x) \neq f(y)$. So we have that $A \preceq \wp(A)$.

We present the slow proof if sec is present, otherwise a faster proof matching sec.

□